AI & COMPUTER-VISION BASED SURFACE & PRODUCT DEFECT INSPECTION

Automated Quality Inspection Report

Inspection ID:	SIM-2611012103-00101	**Date & Time:**	2026-11-01 23:57:12
Product / Part:	Metal Nut	**Batch / Lot:**	-
Product Line:	MVTEC REFERENCE	**Station:**	-
Inspector:	Quality Inspector	**Outcome:**	**PASS**

DEFECT SUMMARY

Defect Type	Count	Critical	High	Medium	Low
Discolouration	1	0	0	0	1

DETECTION LOG

#	Type	Severity	Confidence	Location (px)	Detector	Review
1	Discolouration	**LOW**	96%	406,400 - 600,521	YOLO_SEG +ANOMALY	PENDING

ROOT-CAUSE HINTS

Discolouration (1)
• Inconsistent curing, heat treatment, or coating thickness
• Oxidation / corrosion from moisture exposure or contamination
• Raw-material or paint batch variation
→ Check oven / curing temperature uniformity and dwell time
→ Review storage humidity and batch traceability of coating material

DETERMINATION & SIGN-OFF

Inspector Remarks: Automated computer-vision inspection; detections are subject to inspector confirmation.

Analysed by: Defect Inspection Engine (OPENCV)

Quality Sign-off: _____